



2026. "Año del Humanismo Mexicano en el Estado de México"

Tecnológico de estudios Superiores de Villa Guerrero.

Ingeniería en Sistemas Computacionales

UNIDAD DE APRENDIZAJE:

SISTEMAS OPERATIVOS

PRÁCTICA:

PRACTICA 1. Explorando procesos con ps en Linux

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INTRODUCCION

En esta práctica trabajamos con el comando "ps" en un sistema Linux, con el propósito de conocer cómo se pueden visualizar y consultar los procesos que se encuentran activos en el sistema. Los procesos son las tareas que se están ejecutando y conocer su información nos permite tener una mejor idea de lo que está ocurriendo dentro del sistema operativo.

Durante la práctica utilizamos diferentes formas del comando ps para observar los procesos de distintas maneras. También realizamos filtros y ordenamientos para obtener información más específica, como el identificador de cada proceso (PID), el nombre del comando y el porcentaje de uso de CPU. De esta manera, pudimos observar que un mismo comando puede utilizarse de diferentes formas dependiendo de la información que necesitemos consultar.

El desarrollo de esta práctica nos permitió familiarizarnos con la terminal de Linux y comprender de una manera más práctica cómo se pueden consultar y analizar los procesos activos mediante comandos.

OBJETIVO

Aprender a usar el comando ps para visualizar, filtrar y entender los procesos activos en el sistema.

REQUISITOS

- Sistema Linux (Debian, Ubuntu, Arch, etc.)
- Terminal activa





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DESARROLLO

Para realizar esta práctica, trabajamos con diferentes opciones del comando `ps` en Linux, utilizando la terminal para consultar información sobre los procesos que se encuentran activos en el sistema. En cada apartado utilizamos un comando distinto para observar la información que proporciona y comprender cómo podemos consultarla de acuerdo con nuestras necesidades.

1. Visualización de los procesos del usuario actual

Primero, utilizamos el comando `ps`, el cual permite mostrar los procesos que se están ejecutando en la sesión actual del usuario. Al ejecutarlo en la terminal, podemos observar información de los procesos, como su identificador (PID), la terminal asociada y el tiempo de ejecución, entre otros datos. Esto nos permite conocer algunos de los procesos que se encuentran activos.

Comando utilizado: `ps`

Ilustración 1 comando `ps`

```
user@debian:~$ ps
  PID TTY          TIME CMD
  4638 pts/0        00:00 bash
  4652 pts/0        00:00  ps
user@debian:~$
```



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2. Visualización de los procesos con formato extendido

Después, utilizamos el comando `ps -ef`, que permite consultar los procesos del sistema con información más detallada. A diferencia del comando anterior, esta opción muestra una lista más amplia de procesos e incluye datos como el usuario que los ejecuta, el identificador del proceso (PID), el identificador del proceso padre (PPID) y el comando correspondiente. Esto facilita la consulta de los procesos que se encuentran en ejecución.

Comando utilizado: `ps -ef`

Ilustración 2 comando `ps -ef`

```
user@debian:~$ ps -ef
UID          PID    PPID  C  STIME TTY          TIME CMD
root           1      0  0  22:59 ?        00:00:01 /sbin/init components splash
root           2      0  0  22:59 ?        00:00:00 [kthreadd]
root           3      2  0  22:59 ?        00:00:00 [pool_workqueue_release]
root           4      2  0  22:59 ?        00:00:00 [kworker/R-kvfree_rcu_reclaim]
root           5      2  0  22:59 ?        00:00:00 [kworker/R-rcu_gp]
root           6      2  0  22:59 ?        00:00:00 [kworker/R-sync_wq]
root           7      2  0  22:59 ?        00:00:00 [kworker/R-slub_flushwq]
root           8      2  0  22:59 ?        00:00:00 [kworker/R-netns]
root          10      2  0  22:59 ?        00:00:00 [kworker/0:1-events]
root          11      2  0  22:59 ?        00:00:00 [kworker/0:0H-events_highpri]
root          13      2  0  22:59 ?        00:00:00 [kworker/R-mm_percpu_wq]
root          14      2  0  22:59 ?        00:00:00 [rcu_tasks_kthread]
root          15      2  0  22:59 ?        00:00:00 [rcu_tasks_rude_kthread]
root          16      2  0  22:59 ?        00:00:00 [rcu_tasks_trace_kthread]
root          17      2  0  22:59 ?        00:00:00 [ksoftirqd/0]
root          18      2  0  22:59 ?        00:00:00 [rcu_preempt]
root          19      2  0  22:59 ?        00:00:00 [rcu_exp_par_gp_kthread_worker/0]
root          20      2  0  22:59 ?        00:00:00 [rcu_exp_gp_kthread_worker]
root          21      2  0  22:59 ?        00:00:00 [migration/0]
root          22      2  0  22:59 ?        00:00:00 [idle_inject/0]
root          23      2  0  22:59 ?        00:00:00 [cpuhp/0]
root          24      2  0  22:59 ?        00:00:00 [cpuhp/1]
root          25      2  0  22:59 ?        00:00:00 [idle_inject/1]
root          26      2  0  22:59 ?        00:00:00 [migration/1]
root          27      2  0  22:59 ?        00:00:00 [ksoftirqd/1]
root          28      2  0  22:59 ?        00:00:00 [kworker/1:0-events]
root          29      2  0  22:59 ?        00:00:00 [kworker/1:0H-events_highpri]
root          30      2  0  22:59 ?        00:00:00 [cpuhp/2]
root          31      2  0  22:59 ?        00:00:00 [idle_inject/2]
root          32      2  0  22:59 ?        00:00:00 [migration/2]
root          33      2  0  22:59 ?        00:00:00 [ksoftirqd/2]
root          35      2  0  22:59 ?        00:00:00 [kworker/2:0H-events_highpri]
root          36      2  0  22:59 ?        00:00:00 [cpuhp/3]
root          37      2  0  22:59 ?        00:00:00 [idle_inject/3]
root          38      2  0  22:59 ?        00:00:00 [migration/3]
root          39      2  0  22:59 ?        00:00:00 [ksoftirqd/3]
root          41      2  0  22:59 ?        00:00:00 [kworker/3:0H-events_highpri]
root          42      2  0  22:59 ?        00:00:00 [cpuhp/4]
root          43      2  0  22:59 ?        00:00:00 [idle_inject/4]
root          44      2  0  22:59 ?        00:00:00 [migration/4]
root          45      2  0  22:59 ?        00:00:00 [ksoftirqd/4]
root          47      2  0  22:59 ?        00:00:00 [kworker/4:0H-events_highpri]
root          48      2  0  22:59 ?        00:00:00 [cpuhp/5]
root          49      2  0  22:59 ?        00:00:00 [idle_inject/5]
root          50      2  0  22:59 ?        00:00:00 [migration/5]
root          51      2  0  22:59 ?        00:00:00 [ksoftirqd/5]
root          52      2  0  22:59 ?        00:00:00 [kworker/5:0-events]
root          53      2  0  22:59 ?        00:00:00 [kworker/5:0H-events_highpri]
root          54      2  0  22:59 ?        00:00:00 [cpuhp/6]
root          55      2  0  22:59 ?        00:00:00 [idle_inject/6]
root          56      2  0  22:59 ?        00:00:00 [migration/6]
root          57      2  0  22:59 ?        00:00:00 [ksoftirqd/6]
root          59      2  0  22:59 ?        00:00:00 [kworker/6:0H-events_highpri]
root          60      2  0  22:59 ?        00:00:00 [cpuhp/7]
root          61      2  0  22:59 ?        00:00:00 [idle_inject/7]
root          62      2  0  22:59 ?        00:00:00 [migration/7]
root          63      2  0  22:59 ?        00:00:00 [ksoftirqd/7]
root          65      2  0  22:59 ?        00:00:00 [kworker/7:0H-events_highpri]
root          66      2  0  22:59 ?        00:00:00 [cpuhp/8]
root          67      2  0  22:59 ?        00:00:00 [idle_inject/8]
```




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root	68	2	0 22:59 ?	00:00:00	[migration/8]
root	69	2	0 22:59 ?	00:00:00	[ksoftirqd/8]
root	70	2	0 22:59 ?	00:00:00	[kworker/8:0-lub_flushwq]
root	71	2	0 22:59 ?	00:00:00	[kworker/8:0H-events_highpri]
root	72	2	0 22:59 ?	00:00:00	[cpuhp/9]
root	73	2	0 22:59 ?	00:00:00	[idle_inject/9]
root	74	2	0 22:59 ?	00:00:00	[migration/9]
root	75	2	0 22:59 ?	00:00:00	[ksoftirqd/9]
root	76	2	0 22:59 ?	00:00:00	[kworker/9:0-cgroup_free]
root	77	2	0 22:59 ?	00:00:00	[kworker/9:0H-events_highpri]
root	78	2	0 22:59 ?	00:00:00	[cpuhp/10]
root	79	2	0 22:59 ?	00:00:00	[idle_inject/10]
root	80	2	0 22:59 ?	00:00:00	[migration/10]
root	81	2	0 22:59 ?	00:00:00	[ksoftirqd/10]
root	83	2	0 22:59 ?	00:00:00	[kworker/10:0H-events_highpri]
root	84	2	0 22:59 ?	00:00:00	[cpuhp/11]
root	85	2	0 22:59 ?	00:00:00	[idle_inject/11]
root	86	2	0 22:59 ?	00:00:00	[migration/11]
root	87	2	0 22:59 ?	00:00:00	[ksoftirqd/11]
root	89	2	0 22:59 ?	00:00:00	[kworker/11:0H-events_highpri]
root	90	2	0 22:59 ?	00:00:00	[cpuhp/12]
root	91	2	0 22:59 ?	00:00:00	[idle_inject/12]
root	92	2	0 22:59 ?	00:00:00	[migration/12]
root	93	2	0 22:59 ?	00:00:00	[ksoftirqd/12]
root	95	2	0 22:59 ?	00:00:00	[kworker/12:0H-events_highpri]
root	96	2	0 22:59 ?	00:00:00	[cpuhp/13]
root	97	2	0 22:59 ?	00:00:00	[idle_inject/13]
root	98	2	0 22:59 ?	00:00:00	[migration/13]
root	99	2	0 22:59 ?	00:00:00	[ksoftirqd/13]
root	101	2	0 22:59 ?	00:00:00	[kworker/13:0H-events_highpri]
root	102	2	0 22:59 ?	00:00:00	[cpuhp/14]
root	103	2	0 22:59 ?	00:00:00	[idle_inject/14]
root	104	2	0 22:59 ?	00:00:00	[migration/14]
root	105	2	0 22:59 ?	00:00:00	[ksoftirqd/14]
root	106	2	0 22:59 ?	00:00:00	[kworker/14:0-mm_percpu_wq]
root	107	2	0 22:59 ?	00:00:00	[kworker/14:0H-events_highpri]
root	108	2	0 22:59 ?	00:00:00	[cpuhp/15]
root	109	2	0 22:59 ?	00:00:00	[idle_inject/15]
root	110	2	0 22:59 ?	00:00:00	[migration/15]
root	111	2	0 22:59 ?	00:00:00	[ksoftirqd/15]
root	113	2	0 22:59 ?	00:00:00	[kworker/15:0H-events_highpri]
root	115	2	0 22:59 ?	00:00:00	[kworker/u64:1-events_unbound]
root	118	2	0 22:59 ?	00:00:00	[kworker/u64:4-kvfree_rcu_reclaim]
root	121	2	0 22:59 ?	00:00:00	[kworker/u64:7-gfx_0.0.0]
root	122	2	0 22:59 ?	00:00:00	[kworker/u64:8-events_unbound]
root	128	2	0 22:59 ?	00:00:00	[kworker/u64:14-events_unbound]
root	130	2	0 22:59 ?	00:00:00	[kdevtmpfs]
root	131	2	0 22:59 ?	00:00:00	[kworker/R-inet_frag_wq]
root	132	2	0 22:59 ?	00:00:00	[kauditd]
root	133	2	0 22:59 ?	00:00:00	[khungtaskd]
root	134	2	0 22:59 ?	00:00:00	[oom_reaper]
root	135	2	0 22:59 ?	00:00:00	[kworker/R-writeback]
root	136	2	0 22:59 ?	00:00:00	[kcompactd0]
root	137	2	0 22:59 ?	00:00:00	[ksmd]
root	138	2	0 22:59 ?	00:00:00	[khugepaged]
root	139	2	0 22:59 ?	00:00:00	[kworker/R-kintegrityd]
root	140	2	0 22:59 ?	00:00:00	[kworker/R-kblockd]
root	141	2	0 22:59 ?	00:00:00	[kworker/R-blkcq_punt_bio]
root	142	2	0 22:59 ?	00:00:00	[irq/9-acpi]
root	145	2	0 22:59 ?	00:00:00	[kworker/R-tpm_dev_wq]
root	146	2	0 22:59 ?	00:00:00	[kworker/R-edac_poller]
root	147	2	0 22:59 ?	00:00:00	[kworker/R-devfreq_wq]

root	148	2	0 22:59 ?	00:00:00	[kworker/R-quota_events_unbound]
root	149	2	0 22:59 ?	00:00:00	[kworker/0:1H-kblockd]
root	150	2	0 22:59 ?	00:00:00	[irq/29-AMD-Vi]
root	151	2	0 22:59 ?	00:00:00	[kswapd0]
root	153	2	0 22:59 ?	00:00:00	[kworker/15:1-events]
root	160	2	0 22:59 ?	00:00:00	[kworker/R-kthrottd]
root	165	2	0 22:59 ?	00:00:00	[kworker/R-acpi_thermal_pm]
root	166	2	0 22:59 ?	00:00:00	[kworker/4:1-events]
root	167	2	0 22:59 ?	00:00:00	[hwrng]
root	168	2	0 22:59 ?	00:00:00	[kworker/R-nd]
root	169	2	0 22:59 ?	00:00:00	[kworker/R-nd]
root	170	2	0 22:59 ?	00:00:00	[kworker/R-nd]
root	173	2	0 22:59 ?	00:00:00	[kworker/4:1H-kblockd]
root	176	2	0 22:59 ?	00:00:00	[kworker/4:1H-kblockd]
root	178	2	0 22:59 ?	00:00:00	[kworker/u65:0-ttm]
root	226	2	0 22:59 ?	00:00:00	[irq/30-ACPI-Event]
root	247	2	0 22:59 ?	00:00:00	[kworker/1:1H-kblockd]
root	248	2	0 22:59 ?	00:00:00	[irq/31-ACPI-Event]
root	249	2	0 22:59 ?	00:00:00	[irq/32-ACPI-Event]
root	250	2	0 22:59 ?	00:00:00	[irq/33-ACPI-Event]
root	251	2	0 22:59 ?	00:00:00	[irq/34-ACPI-Event]
root	252	2	0 22:59 ?	00:00:00	[kworker/3:1-cgroup_free]
root	253	2	0 22:59 ?	00:00:00	[irq/35-ACPI-Event]
root	254	2	0 22:59 ?	00:00:00	[kworker/9:1-mm_percpu_wq]
root	255	2	0 22:59 ?	00:00:00	[kworker/6:1-cgroup_free]
root	256	2	0 22:59 ?	00:00:00	[kworker/7:1-mm_percpu_wq]
root	285	2	0 22:59 ?	00:00:00	[kworker/2:1H-kblockd]
root	286	2	0 22:59 ?	00:00:00	[kworker/5:1H-kblockd]
root	297	2	0 22:59 ?	00:00:00	[kworker/13:1H-kblockd]
root	309	2	0 22:59 ?	00:00:00	[kworker/7:1H-kblockd]
root	310	2	0 22:59 ?	00:00:00	[kworker/15:1H-kblockd]
root	311	2	0 22:59 ?	00:00:00	[kworker/14:1H-kblockd]
root	312	2	0 22:59 ?	00:00:00	[kworker/8:1H-kblockd]
root	313	2	0 22:59 ?	00:00:00	[kworker/6:1H-kblockd]
root	314	2	0 22:59 ?	00:00:00	[kworker/3:1H-kblockd]
root	315	2	0 22:59 ?	00:00:00	[kworker/12:1H-kblockd]
root	316	2	0 22:59 ?	00:00:00	[kworker/10:1H-kblockd]
root	317	2	0 22:59 ?	00:00:00	[kworker/9:1H-kblockd]
root	319	2	0 22:59 ?	00:00:00	[kworker/11:1H-kblockd]
root	340	2	0 22:59 ?	00:00:00	[kworker/15:2-mm_percpu_wq]
root	351	2	0 22:59 ?	00:00:00	[kworker/R-nvme-wq]
root	352	2	0 22:59 ?	00:00:00	[kworker/R-nvme-reset-wq]
root	354	2	0 22:59 ?	00:00:00	[kworker/R-nvme-auth-wq]
root	355	2	0 22:59 ?	00:00:00	[watchdog]
root	356	2	0 22:59 ?	00:00:00	[kworker/11:1-cgroup_free]
root	357	2	0 22:59 ?	00:00:00	[irq/62-PNP0C50:00]
root	358	2	0 22:59 ?	00:00:00	[kworker/10:1-mm_percpu_wq]
root	361	2	0 22:59 ?	00:00:00	[kworker/2:4-events_freezable_pwr_ef]
root	363	2	0 22:59 ?	00:00:00	[kworker/2:6-events]
root	365	2	0 22:59 ?	00:00:00	[kworker/1:2-events]
root	367	2	0 22:59 ?	00:00:00	[kworker/3:3-events]
root	370	2	0 22:59 ?	00:00:00	[kworker/13:1-mm_percpu_wq]
root	371	2	0 22:59 ?	00:00:00	[kworker/13:2-mm_percpu_wq]
root	372	2	0 22:59 ?	00:00:00	[kworker/R-andpup-reset-dev]
root	373	2	0 22:59 ?	00:00:00	[kworker/R-ttm]
root	374	2	0 22:59 ?	00:00:00	[kworker/12:1-events]
root	376	2	0 22:59 ?	00:00:00	[kworker/6:2-mm_percpu_wq]
root	377	2	0 22:59 ?	00:00:00	[scsi_ah_0]
root	378	2	0 22:59 ?	00:00:00	[kworker/R-scsi_tmf_0]
root	379	2	0 22:59 ?	00:00:00	[usb-storage]
root	380	2	0 22:59 ?	00:00:00	[kworker/R-uas]

root	148	2	0 22:59 ?	00:00:00	[kworker/R-quota_events_unbound]
root	149	2	0 22:59 ?	00:00:00	[kworker/0:1H-kblockd]
root	150	2	0 22:59 ?	00:00:00	[irq/29-AMD-Vi]
root	151	2	0 22:59 ?	00:00:00	[kswapd0]
root	153	2	0 22:59 ?	00:00:00	[kworker/15:1-events]
root	159	2	0 22:59 ?	00:00:00	[kworker/R-kthrotld]
root	165	2	0 22:59 ?	00:00:00	[kworker/R-acpi_thermal_pm]
root	166	2	0 22:59 ?	00:00:00	[kworker/4:1-events]
root	167	2	0 22:59 ?	00:00:00	[hwrng]
root	168	2	0 22:59 ?	00:00:00	[kworker/4:2-mm_percpu_wq]
root	169	2	0 22:59 ?	00:00:00	[kworker/R-nd]
root	170	2	0 22:59 ?	00:00:00	[kworker/R-nd]
root	173	2	0 22:59 ?	00:00:00	[kworker/4:1H-kblockd]
root	176	2	0 22:59 ?	00:00:00	[kworker/R-kstrp]
root	178	2	0 22:59 ?	00:00:00	[kworker/u65:0-ttm]
root	246	2	0 22:59 ?	00:00:00	[irq/30-ACPI-Event]
root	247	2	0 22:59 ?	00:00:00	[kworker/1:1H-kblockd]
root	248	2	0 22:59 ?	00:00:00	[irq/31-ACPI-Event]
root	249	2	0 22:59 ?	00:00:00	[irq/32-ACPI-Event]
root	250	2	0 22:59 ?	00:00:00	[irq/33-ACPI-Event]
root	251	2	0 22:59 ?	00:00:00	[irq/34-ACPI-Event]
root	252	2	0 22:59 ?	00:00:00	[kworker/3:1-cgroup_free]
root	253	2	0 22:59 ?	00:00:00	[irq/35-ACPI-Event]
root	254	2	0 22:59 ?	00:00:00	[kworker/9:1-mm_percpu_wq]
root	255	2	0 22:59 ?	00:00:00	[kworker/6:1-cgroup_free]
root	256	2	0 22:59 ?	00:00:00	[kworker/7:1-mm_percpu_wq]
root	285	2	0 22:59 ?	00:00:00	[kworker/2:1H-kblockd]
root	286	2	0 22:59 ?	00:00:00	[kworker/5:1H-kblockd]
root	297	2	0 22:59 ?	00:00:00	[kworker/13:1H-kblockd]
root	309	2	0 22:59 ?	00:00:00	[kworker/7:1H-kblockd]
root	310	2	0 22:59 ?	00:00:00	[kworker/15:1H-kblockd]
root	311	2	0 22:59 ?	00:00:00	[kworker/14:1H-kblockd]
root	312	2	0 22:59 ?	00:00:00	[kworker/8:1H-kblockd]
root	313	2	0 22:59 ?	00:00:00	[kworker/6:1H-kblockd]
root	314	2	0 22:59 ?	00:00:00	[kworker/3:1H-kblockd]
root	315	2	0 22:59 ?	00:00:00	[kworker/12:1H-kblockd]
root	316	2	0 22:59 ?	00:00:00	[kworker/10:1H-kblockd]
root	317	2	0 22:59 ?	00:00:00	[kworker/9:1H-kblockd]
root	319	2	0 22:59 ?	00:00:00	[kworker/11:1H-kblockd]
root	348	2	0 22:59 ?	00:00:00	[kworker/15:2-mm_percpu_wq]
root	351	2	0 22:59 ?	00:00:00	[kworker/R-nvme-wq]
root	352	2	0 22:59 ?	00:00:00	[kworker/R-nvme-reset-wq]
root	353	2	0 22:59 ?	00:00:00	[kworker/R-nvme-delete-wq]
root	354	2	0 22:59 ?	00:00:00	[kworker/R-nvme-auth-wq]
root	355	2	0 22:59 ?	00:00:00	[kworker/1:1-cgroup_free]
root	356	2	0 22:59 ?	00:00:00	[kworker/16:1-cgroup_free]
root	357	2	0 22:59 ?	00:00:00	[irq/62-PNP0C80]
root	358	2	0 22:59 ?	00:00:00	[kworker/10:1-mm_percpu_wq]
root	361	2	0 22:59 ?	00:00:00	[kworker/2:4-events_freezable_pwr_e
root	363	2	0 22:59 ?	00:00:00	[kworker/2:6-events]
root	372	2	0 22:59 ?	00:00:00	[kworker/3-events]
root	376	2	0 22:59 ?	00:00:00	[kworker/3:3-events]
root	370	2	0 22:59 ?	00:00:00	[kworker/13:1-mm_percpu_wq]
root	371	2	0 22:59 ?	00:00:00	[kworker/13:2-mm_percpu_wq]
root	372	2	0 22:59 ?	00:00:00	[kworker/R-andcpu-reset-dev]
root	373	2	0 22:59 ?	00:00:00	[kworker/1-events]
root	374	2	0 22:59 ?	00:00:00	[kworker/12:1-events]
root	376	2	0 22:59 ?	00:00:00	[kworker/6:2-mm_percpu_wq]
root	377	2	0 22:59 ?	00:00:00	[scsi eh 0]
root	378	2	0 22:59 ?	00:00:00	[kworker/R-scsi_tmf_0]
root	379	2	0 22:59 ?	00:00:00	[usb-storage]
root	380	2	0 22:59 ?	00:00:00	[kworker/2:5]



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3. Filtrado de procesos

En el siguiente apartado, utilizamos el comando `ps -ef | grep firefox` para buscar procesos relacionados con Firefox. Para ello, combinamos el comando `ps -ef` con `grep`, que permite filtrar la información y mostrar las líneas que coinciden con el texto indicado. De esta manera, podemos localizar con mayor facilidad los procesos relacionados con un programa específico.

Comando utilizado: `ps -ef | grep firefox`

Ilustración 3 comando `ps -ef | grep firefox`

```
user@debian:~$ ps -ef | grep firefox
user      2839   1914 13 22:59 ?        00:01:48 /usr/lib/firefox-esr/firefox-esr
user      2869   1736  0 22:59 ?        00:00:00 /usr/lib/firefox-esr/crashhelper 2839 9 /tmp/ 11
user      2943   2839  0 22:59 ?        00:00:00 /usr/lib/firefox-esr/firefox-esr -contentproc -parentBuildID 20260609153453 -prefsHandle 0:26640 -prefMapHandle 1:268799 -sandboxReporter 2 -chrootClient 3 -ip
cHandle 4 -initialChannelId {e4edef9e-288d-48b0-9e27-039cc042cd51} -parentPid 2839 -crashReporter 5 -crashHelper 6 -appDir /usr/lib/firefox-esr/browser 1 socket
user      3019   2839  0 22:59 ?        00:00:00 /usr/lib/firefox-esr/firefox-esr -contentproc -isForBrowser -prefsHandle 0:28015 -prefMapHandle 1:268799 -jsInitHandle 2:242716 -parentBuildID 20260609153453 -
sandboxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChannelId {914f901b-fd10-45a3-9b02-ca3ce7f5db89} -parentPid 2839 -crashReporter 6 -crashHelper 7 -greomni /usr/lib/firefox-esr/omni.ja -appomni /usr/lib/fir
efox-esr/browser/omni.ja -appDir /usr/lib/firefox-esr/browser 2 tab
user      3024   2839  0 22:59 ?        00:00:01 /usr/lib/firefox-esr/firefox-esr -contentproc -parentBuildID 20260609153453 -prefsHandle 0:28060 -prefMapHandle 1:268799 -sandboxReporter 2 -chrootClient 3 -ip
cHandle 4 -initialChannelId {ce7d73df-7e2b-43bd-8f70-2f2debcbf102} -parentPid 2839 -crashReporter 5 -crashHelper 6 -appDir /usr/lib/firefox-esr/browser 3 rdd
user      3058   2839  0 22:59 ?        00:00:02 /usr/lib/firefox-esr/firefox-esr -contentproc -isForBrowser -prefsHandle 0:28136 -prefMapHandle 1:268799 -jsInitHandle 2:242716 -parentBuildID 20260609153453 -
sandboxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChannelId {060bd64e-8fb5-4d19-bd73-b3f9afb143d3} -parentPid 2839 -crashReporter 6 -crashHelper 7 -greomni /usr/lib/firefox-esr/omni.ja -appomni /usr/lib/fir
efox-esr/browser/omni.ja -appDir /usr/lib/firefox-esr/browser 4 tab
user      3170   2839  0 22:59 ?        00:00:07 /usr/lib/firefox-esr/firefox-esr -contentproc -parentBuildID 20260609153453 -sandboxingKind 0 -prefsHandle 0:40107 -prefMapHandle 1:268799 -sandboxReporter 2 -
chrootClient 3 -ipcHandle 4 -initialChannelId {592c8a4c-aa96-4314-bd7f-5aed05639617} -parentPid 2839 -crashReporter 5 -crashHelper 6 -appDir /usr/lib/firefox-esr/browser 6 utility
user      3196   2839 25 22:59 ?        00:03:21 /usr/lib/firefox-esr/firefox-esr -contentproc -isForBrowser -prefsHandle 0:38293 -prefMapHandle 1:268799 -jsInitHandle 2:242716 -parentBuildID 20260609153453 -
sandboxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChannelId {55031904-0f5a-44ba-997c-14bca8c4e4d3} -parentPid 2839 -crashReporter 6 -crashHelper 7 -greomni /usr/lib/firefox-esr/omni.ja -appomni /usr/lib/fir
efox-esr/browser/omni.ja -appDir /usr/lib/firefox-esr/browser 9 tab
user      3558   2839  0 23:01 ?        00:00:00 /usr/lib/firefox-esr/firefox-esr -contentproc -isForBrowser -prefsHandle 0:38673 -prefMapHandle 1:268799 -jsInitHandle 2:242716 -parentBuildID 20260609153453 -
sandboxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChannelId {b6d03538-3d5c-4ad9-9510-70d1d382c07d} -parentPid 2839 -crashReporter 6 -crashHelper 7 -greomni /usr/lib/firefox-esr/omni.ja -appomni /usr/lib/fir
efox-esr/browser/omni.ja -appDir /usr/lib/firefox-esr/browser 12 tab
user      3821   2839  8 23:01 ?        00:00:55 /usr/lib/firefox-esr/firefox-esr -contentproc -isForBrowser -prefsHandle 0:38673 -prefMapHandle 1:268799 -jsInitHandle 2:242716 -parentBuildID 20260609153453 -
sandboxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChannelId {c53fc003-f628-4121-8053-a0a392f680cc} -parentPid 2839 -crashReporter 6 -crashHelper 7 -greomni /usr/lib/firefox-esr/omni.ja -appomni /usr/lib/fir
efox-esr/browser/omni.ja -appDir /usr/lib/firefox-esr/browser 15 tab
user      3916   2839  0 23:01 ?        00:00:00 /usr/lib/firefox-esr/firefox-esr -contentproc -isForBrowser -prefsHandle 0:38673 -prefMapHandle 1:268799 -jsInitHandle 2:242716 -parentBuildID 20260609153453 -
sandboxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChannelId {0e1036d8-6e4a-4a5e-a800-07fd52a58b3b} -parentPid 2839 -crashReporter 6 -crashHelper 7 -greomni /usr/lib/firefox-esr/omni.ja -appomni /usr/lib/fir
efox-esr/browser/omni.ja -appDir /usr/lib/firefox-esr/browser 17 tab
user      4355   2839  0 23:02 ?        00:00:00 /usr/lib/firefox-esr/firefox-esr -contentproc -isForBrowser -prefsHandle 0:38760 -prefMapHandle 1:268799 -jsInitHandle 2:242716 -parentBuildID 20260609153453 -
sandboxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChannelId {8036dbf5-3f3d-4aed-8b6f-000dfa0429af} -parentPid 2839 -crashReporter 6 -crashHelper 7 -greomni /usr/lib/firefox-esr/omni.ja -appomni /usr/lib/fir
efox-esr/browser/omni.ja -appDir /usr/lib/firefox-esr/browser 19 tab
user      4392   2839  0 23:02 ?        00:00:00 /usr/lib/firefox-esr/firefox-esr -contentproc -isForBrowser -prefsHandle 0:38760 -prefMapHandle 1:268799 -jsInitHandle 2:242716 -parentBuildID 20260609153453 -
sandboxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChannelId {82dd2653-60da-4922-9589-2cca0a2bc3e7} -parentPid 2839 -crashReporter 6 -crashHelper 7 -greomni /usr/lib/firefox-esr/omni.ja -appomni /usr/lib/fir
efox-esr/browser/omni.ja -appDir /usr/lib/firefox-esr/browser 20 tab
user      4480   2839  0 23:02 ?        00:00:00 /usr/lib/firefox-esr/firefox-esr -contentproc -isForBrowser -prefsHandle 0:38760 -prefMapHandle 1:268799 -jsInitHandle 2:242716 -parentBuildID 20260609153453 -
sandboxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChannelId {00521cd9-0ee3-46ef-a15b-58694385e8bd} -parentPid 2839 -crashReporter 6 -crashHelper 7 -greomni /usr/lib/firefox-esr/omni.ja -appomni /usr/lib/fir
efox-esr/browser/omni.ja -appDir /usr/lib/firefox-esr/browser 21 tab
user      5436   4638  0 23:12 pts/0    00:00:00 grep firefox
user@debian:~$
```



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4. Visualización de los procesos de un usuario específico

Posteriormente, utilizamos el comando `ps -u nombre_usuario`, que permite consultar los procesos asociados a un usuario determinado. Para utilizarlo, se debe sustituir `nombre_usuario` por el nombre del usuario que se desea consultar. Esta opción resulta útil para identificar qué procesos están relacionados con una cuenta específica dentro del sistema.

Comando utilizado: `ps -u nombre_usuario`

Ilustración 4 comando `ps -u nombre_usuario`

```
user@debian:~$ ps -u user
  PID TTY          TIME CMD
 1736 ?        00:00:00 systemd
 1738 ?        00:00:00 (sd-pam)
 1758 ?        00:00:00 dbus-daemon
 1760 ?        00:00:01 pipewire
 1762 ?        00:00:00 pipewire
 1763 ?        00:00:00 wireplumber
 1764 ?        00:00:02 pipewire-pulse
 1766 ?        00:00:00 gnome-keyring-d
 1767 ?        00:00:00 mpris-proxy
 1782 tty2      00:00:00 gdm-wayland-ses
 1787 tty2      00:00:00 gnome-session-b
 1838 ?        00:00:00 gcr-ssh-agent
 1839 ?        00:00:00 gnome-session-c
 1840 ?        00:00:00 ssh-agent
 1851 ?        00:00:00 gvfsd
 1858 ?        00:00:00 gvfsd-fuse
 1861 ?        00:00:00 gnome-session-b
 1891 ?        00:00:00 at-spi-bus-laun
 1908 ?        00:00:00 dbus-daemon
 1914 ?        00:00:44 gnome-shell
 1939 ?        00:00:00 xdg-permission-
 1977 ?        00:00:00 at-spi2-registr
 2012 ?        00:00:00 gnome-shell-cal
 2022 ?        00:00:00 dconf-service
 2023 ?        00:00:00 evolution-sourc
 2027 ?        00:00:00 gjs
 2047 ?        00:00:01 ibus-daemon
 2048 ?        00:00:00 gsd-ally-settin
 2049 ?        00:00:00 gsd-color
 2051 ?        00:00:00 gsd-datetime
 2052 ?        00:00:00 gsd-housekeepin
 2053 ?        00:00:00 gsd-keyboard
 2055 ?        00:00:00 gsd-media-keys
 2056 ?        00:00:00 gsd-power
 2058 ?        00:00:00 gsd-print-notif
 2060 ?        00:00:00 gsd-rfkill
 2064 ?        00:00:00 gsd-screensaver
 2065 ?        00:00:00 gsd-sharing
 2067 ?        00:00:00 gsd-smartcard
 2068 ?        00:00:00 gsd-sound
 2070 ?        00:00:00 gsd-usb-protect
 2073 ?        00:00:00 gsd-wacom
 2082 ?        00:00:00 gvfs-udisks2-vo
 2128 ?        00:00:00 gjs
 2166 ?        00:00:00 gsd-printer
 2173 ?        00:00:00 gvfs-mtp-volume
 2204 ?        00:00:00 gvfs-gphoto2-vo
 2209 ?        00:00:00 gsd-disk-utilit
 2223 ?        00:00:09 gnome-software
 2230 ?        00:00:00 evolution-alarm
 2285 ?        00:00:00 gvfs-goa-volume
 2296 ?        00:00:00 goa-daemon
 2306 ?        00:00:00 goa-identity-se
 2308 ?        00:00:00 gvfs-afc-volume
 2319 ?        00:00:00 Xwayland
 2392 ?        00:00:00 evolution-calen
 2403 ?        00:00:00 gvfsd-metadata
 2410 ?        00:00:00 evolution-addre
 2435 ?        00:00:00 xdg-desktop-por
 2438 ?        00:00:00 gsd-xsettings
```





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```
2438 ? 00:00:00 gsd-xsettings
2446 ? 00:00:00 xdg-document-po
2470 ? 00:00:00 mutter-x11-fram
2540 ? 00:00:00 ibus-dconf
2541 ? 00:00:01 ibus-extension-
2548 ? 00:00:00 ibus-portal
2550 ? 00:00:00 localsearch-3
2551 ? 00:00:00 xdg-desktop-por
2566 ? 00:00:00 ibus-engine-moz
2574 ? 00:00:00 mozc_server
2584 ? 00:00:00 ibus-x11
2623 ? 00:00:00 xdg-desktop-por
2839 ? 00:01:52 firefox-esr
2869 ? 00:00:00 crashhelper
2943 ? 00:00:00 Socket Process
3019 ? 00:00:00 WebExtensions
3024 ? 00:00:01 RDD Process
3058 ? 00:00:02 Privileged Cont
3170 ? 00:00:08 Utility Process
3196 ? 00:03:35 Isolated Web Co
3298 ? 00:00:00 ibus-engine-sim
3558 ? 00:00:01 Isolated Servic
3821 ? 00:01:00 Isolated Web Co
3916 ? 00:00:00 Isolated Web Co
4355 ? 00:00:00 Web Content
4392 ? 00:00:00 Web Content
4488 ? 00:00:00 Web Content
4514 ? 00:00:01 speech-dispatch
4527 ? 00:00:00 sd_espeak-ng-mb
4528 ? 00:00:00 sd_espeak-ng
4532 ? 00:00:00 sd_dummy
4535 ? 00:00:00 sd_espeak-ng
4559 ? 00:00:01 oosplash
4593 ? 00:00:08 soffice.bin
4630 ? 00:00:05 gnome-terminal-
4638 pts/0 00:00:00 bash
5524 pts/0 00:00:00 ps
user@debian:~$
```



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5. Visualización del identificador y nombre de los procesos

A continuación, utilizamos el comando `ps -eo pid,comm`, que permite mostrar únicamente dos datos de cada proceso: su identificador (PID) y el nombre del comando con el que se ejecuta. Esta opción facilita la lectura de la información, ya que presenta los datos de manera más sencilla y evita mostrar otras columnas que no necesitamos consultar en ese momento.

Comando utilizado: `ps -eo pid,comm`

Ilustración 5 comando `ps -eo pid,comm`

```
user@debian:~$ ps -eo pid,comm
PID COMMAND
1 systemd
2 kthreadd
3 pool_workqueue_release
4 kworker/R-kvfree_rcu_reclaim
5 kworker/R-rcu_gp
6 kworker/R-sync_wq
7 kworker/R-slab_flushwq
8 kworker/R-netns
10 kworker/0:1-events
11 kworker/0:0H-events_highpri
13 kworker/R-mm_percpu_wq
14 rcu_tasks_kthread
15 rcu_tasks_rude_kthread
16 rcu_tasks_trace_kthread
17 ksoftirqd/0
18 rcu_preempt
19 rcu_exp_par_gp_kthread_worker/0
20 rcu_exp_gp_kthread_worker
21 migration/0
22 idle_inject/0
23 cpuhp/0
24 cpuhp/1
25 idle_inject/1
26 migration/1
27 ksoftirqd/1
29 kworker/1:0H-events_highpri
30 cpuhp/2
31 idle_inject/2
32 migration/2
33 ksoftirqd/2
35 kworker/2:0H-events_highpri
36 cpuhp/3
37 idle_inject/3
38 migration/3
39 ksoftirqd/3
41 kworker/3:0H-events_highpri
42 cpuhp/4
43 idle_inject/4
44 migration/4
45 ksoftirqd/4
47 kworker/4:0H-events_highpri
48 cpuhp/5
49 idle_inject/5
50 migration/5
51 ksoftirqd/5
52 kworker/5:0-events
53 kworker/5:0H-events_highpri
54 cpuhp/6
55 idle_inject/6
56 migration/6
57 ksoftirqd/6
59 kworker/6:0H-events_highpri
60 cpuhp/7
61 idle_inject/7
62 migration/7
63 ksoftirqd/7
65 kworker/7:0H-events_highpri
66 cpuhp/8
67 idle_inject/8
68 migration/8
```



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```
68 migration/8
69 ksoftirqd/8
70 kworker/8:0-slab_flushwq
71 kworker/8:0H-events_highpri
72 cpuhp/9
73 idle_inject/9
74 migration/9
75 ksoftirqd/9
77 kworker/9:0H-events_highpri
78 cpuhp/10
79 idle_inject/10
80 migration/10
81 ksoftirqd/10
83 kworker/10:0H-events_highpri
84 cpuhp/11
85 idle_inject/11
86 migration/11
87 ksoftirqd/11
89 kworker/11:0H-events_highpri
90 cpuhp/12
91 idle_inject/12
92 migration/12
93 ksoftirqd/12
95 kworker/12:0H-events_highpri
96 cpuhp/13
97 idle_inject/13
98 migration/13
99 ksoftirqd/13
101 kworker/13:0H-events_highpri
102 cpuhp/14
103 idle_inject/14
104 migration/14
105 ksoftirqd/14
106 kworker/14:0-mm_percpu_wq
107 kworker/14:0H-events_highpri
108 cpuhp/15
109 idle_inject/15
110 migration/15
111 ksoftirqd/15
113 kworker/15:0H-events_highpri
121 kworker/u64:7-kvfree_rcu_reclaim
130 kdevtmpfs
131 kworker/R-inet_frag_wq
132 kauditd
133 khungtaskd
134 oom_reaper
135 kworker/R-writeback
136 kcompactd0
137 ksm
138 khugepaged
139 kworker/R-kintegrityd
140 kworker/R-kblockd
141 kworker/R-blkcg_punt_bio
142 irq/9-acpi
145 kworker/R-tpm_dev_wq
146 kworker/R-edac-poller
147 kworker/R-devfreq_wq
148 kworker/R-quota_events_unbound
149 kworker/0:1H-kblockd
150 irq/29-AMD-Vi
151 kswapd0
153 kworker/15:1-events
154 kworker/R-kthrotld
```

```
153 kworker/15:1-events
160 kworker/R-kthrotld
165 kworker/R-acpi_thermal_pm
166 kworker/4:1-events
167 hwrng
168 kworker/4:2-events
169 kworker/R-mld
170 kworker/R-ipv6_addrconf
173 kworker/4:1H-kblockd
176 kworker/R-kstrp
178 kworker/u65:0-ttm
246 irq/30-ACPI:Event
247 kworker/1:1H-kblockd
248 irq/31-ACPI:Event
249 irq/32-ACPI:Event
250 irq/33-ACPI:Event
251 irq/34-ACPI:Event
252 kworker/3:1-mm_percpu_wq
253 irq/35-ACPI:Event
254 kworker/9:1-events
255 kworker/6:1-cgroup_free
256 kworker/7:1-events
285 kworker/2:1H-kblockd
286 kworker/5:1H-kblockd
297 kworker/13:1H-kblockd
309 kworker/7:1H-kblockd
310 kworker/15:1H-kblockd
311 kworker/14:1H-kblockd
312 kworker/8:1H-kblockd
313 kworker/6:1H-kblockd
314 kworker/3:1H-kblockd
315 kworker/12:1H-kblockd
316 kworker/10:1H-kblockd
317 kworker/9:1H-kblockd
319 kworker/11:1H-kblockd
348 kworker/15:2-mm_percpu_wq
351 kworker/R-nvme-wq
352 kworker/R-nvme-reset-wq
353 kworker/R-nvme-delete-wq
354 kworker/R-nvme-auth-wq
355 watchdogd
357 irq/62-PNP0C50:00
358 kworker/10:1-events
361 kworker/2:4-events_freezable_pwr_efficient
365 kworker/1:2-cgroup_free
371 kworker/13:2-mm_percpu_wq
372 kworker/R-amdgpu-reset-dev
373 kworker/R-ttm
374 kworker/12:1-events
376 kworker/6:2-mm_percpu_wq
377 scsi_ah_0
378 kworker/R-scsi_tmf_0
379 usb-storage
380 kworker/R-uas
381 kworker/R-amdgpu_dm_hpd_rx_offload_wq
382 kworker/R-amdgpu_dm_hpd_rx_offload_wq
383 kworker/R-amdgpu_dm_hpd_rx_offload_wq
384 kworker/R-amdgpu_dm_hpd_rx_offload_wq
385 kworker/R-amdgpu_dm_hpd_rx_offload_wq
386 kworker/R-amdgpu_dm_hpd_rx_offload_wq
387 kworker/R-amdgpu_dm_hpd_rx_offload_wq
388 kworker/R-amdgpu_dm_hpd_rx_offload_wq
```





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```
389 kworker/R-amdgpu_dm_hpd_rx_offload_wq
390 kworker/R-dm_vblank_control_workqueue
391 kworker/R-amdgpu_dm_hpd_wq
392 card0-crtc0
393 card0-crtc1
394 card0-crtc2
395 card0-crtc3
396 kworker/R-gfx_0.0.0
397 kworker/R-comp_1.0.0
398 kworker/R-comp_1.1.0
399 kworker/R-comp_1.2.0
400 kworker/R-comp_1.3.0
401 kworker/R-comp_1.0.1
402 kworker/R-comp_1.1.1
403 kworker/R-comp_1.2.1
404 kworker/R-comp_1.3.1
405 kworker/R-sdma0
406 kworker/R-vcn_unified_0
407 kworker/R-jpeg_dec
408 kworker/u65:1-ttm
417 kworker/R-md
418 kworker/R-md_bitmap
424 kworker/R-raid5wq
606 psimon
620 systemd-journal
674 systemd-udev
679 psimon
733 kworker/10:2-events
760 kworker/R-cryptd
800 kworker/u64:16-kvfree_rcu_reclaim
878 kworker/R-cfg80211
885 kworker/11:2-events
902 kworker/R-kmemstick
903 kworker/u65:2-rtw89_tx_wq
905 kworker/8:2-events_freezable_pwr_efficient
1066 irq/80-rtw89_pci
1587 accounts-daemon
1588 avahi-daemon
1589 bluetoothd
1591 cron
1592 dbus-daemon
1596 low-memory-moni
1597 polkitd
1598 smartd
1599 switcheroo-cont
1601 avahi-daemon
1602 systemd-logind
1603 udisksd
1646 NetworkManager
1647 wpa_supplicant
1655 psimon
1681 ModemManager
1690 krfcomm
1707 cupsd
1721 gdm3
1722 kworker/9:2-mm_percpu_wq
1728 gdm-session-wor
1736 systemd
1738 (sd-pam)
1758 dbus-daemon
1760 pipewire
1762 pipewire
```



```
1760 pipewire
1762 pipewire
1763 wireplumber
1764 pipewire-pulse
1766 gnome-keyring-d
1767 mpris-proxy
1776 rtkit-daemon
1782 gdm-wayland-ses
1787 gnome-session-b
1838 gcr-ssh-agent
1839 gnome-session-c
1840 ssh-agent
1851 gvfsd
1858 gvfsd-fuse
1861 gnome-session-b
1891 at-spi-bus-laun
1908 dbus-daemon
1914 gnome-shell
1939 xdg-permission-
1945 cups-browsed
1977 at-spi2-registr
1978 colord
2012 gnome-shell-cal
2022 dconf-service
2023 evolution-sourc
2027 gjs
2046 upowerd
2047 ibus-daemon
2048 gsd-ally-settin
2049 gsd-color
2051 gsd-datetime
2052 gsd-housekeepin
2053 gsd-keyboard
2055 gsd-media-keys
2056 gsd-power
2058 gsd-print-notif
2060 gsd-rfkill
2064 gsd-screensaver
2065 gsd-sharing
2067 gsd-smartcard
2068 gsd-sound
2070 gsd-usb-protect
2073 gsd-wacom
2082 gvfs-udisks2-vo
2128 gjs
2166 gsd-printer
2173 gvfs-mtp-volume
2204 gvfs-gphoto2-vo
2209 gsd-disk-utilit
2223 gnome-software
2230 evolution-alarm
2285 gvfs-goa-volume
2296 goa-daemon
2306 goa-identity-se
2308 gvfs-afc-volume
2319 Xwayland
2392 evolution-calen
2403 gvfsd-metadata
2410 evolution-addre
2435 xdg-desktop-por
2438 gsd-xsettings
2446 xdg-document-po
2454 fusionmount2
```





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```
2446 xdg-document-po
2454 fusermount3
2470 mutter-x11-fram
2540 ibus-dconf
2541 ibus-extension-
2548 ibus-portal
2550 localsearch-3
2551 xdg-desktop-por
2566 ibus-engine-moz
2574 mozc_server
2584 ibus-x11
2623 xdg-desktop-por
2652 kworker/u65:3-ttm
2653 kworker/u65:4-ttm
2654 kworker/u65:5-ttm
2655 kworker/u65:6-ttm
2656 kworker/u65:7-ttm
2657 kworker/u65:8-ttm
2658 kworker/u65:9-ttm
2659 kworker/u65:10-ttm
2660 kworker/u65:11-ttm
2661 kworker/u65:12-ttm
2662 kworker/u65:13-ttm
2663 kworker/u65:14-ttm
2664 kworker/u65:15-rtw89_tx_wq
2763 fwupd
2839 firefox-esr
2869 crashhelper
2891 kworker/12:2-mm_percpu_wq
2943 Socket Process
2958 power-profiles-
3019 WebExtensions
3024 RDD Process
3058 Privileged Cont
3170 Utility Process
3196 Isolated Web Co
3298 ibus-engine-sim
3558 Isolated Servic
3821 Isolated Web Co
3916 Isolated Web Co
4355 Web Content
4392 Web Content
4488 Web Content
4514 speech-dispatch
4527 sd_espeak-ng-mb
4528 sd_espeak-ng
4532 sd_dummy
4535 sd_espeak-ng
4559 oosplash
4593 soffice.bin
4630 gnome-terminal-
4638 bash
4751 kworker/1:1-mm_percpu_wq
4834 kworker/7:0
4853 kworker/2:0-mm_percpu_wq
4866 kworker/13:0
4987 kworker/3:0-mm_percpu_wq
5163 kworker/14:1-events
5165 kworker/u64:0-kvfree_rcu_reclaim
5194 kworker/11:0
5264 kworker/5:2-events
5292 kworker/0:0
```

```
4866 kworker/13:0
4987 kworker/3:0-mm_percpu_wq
5163 kworker/14:1-events
5165 kworker/u64:0-kvfree_rcu_reclaim
5194 kworker/11:0
5264 kworker/5:2-events
5292 kworker/0:0
5304 kworker/1:0-events_freezable
5312 kworker/u64:1+events_unbound
5317 kworker/4:0-mm_percpu_wq
5703 kworker/u65:16
5722 kworker/5:1
5743 ps
```

```
user@debian:~$
```



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6. Ordenamiento de los procesos según el uso de CPU

Finalmente, utilizamos el comando `ps -eo pid,comm,%cpu --sort=-%cpu`, que permite visualizar el identificador de cada proceso, el nombre del comando y el porcentaje de uso de la CPU. Además, la opción `--sort=-%cpu` organiza los resultados de mayor a menor según el uso del procesador. Esto nos ayuda a identificar qué procesos presentan un mayor consumo de CPU en el momento de realizar la consulta.

Comando utilizado: `ps -eo pid,comm,%cpu --sort=-%cpu`

Ilustración 6 comando `ps -eo pid,comm,%cpu --sort=-%cpu`

```
3/43 ps
user@debian:~$ ps -eo pid,comm,%cpu --sort=-%cpu
PID COMMAND %CPU
3196 Isolated Web Co 22.9
2839 firefox-esr 11.4
3821 Isolated Web Co 8.0
1914 gnome-shell 4.9
3170 Utility Process 1.0
4630 gnome-terminal- 0.9
4593 soffice.bin 0.8
2223 gnome-software 0.7
1066 irq/80-rtw89_pc 0.6
1764 pipewire-pulse 0.2
5924 Web Content 0.2
2763 fwupd 0.2
3058 Privileged Cont 0.2
121 kworker/u64:7-s 0.1
4559 oosplash 0.1
800 kworker/u64:16- 0.1
1760 pipewire 0.1
3024 RDD Process 0.1
5165 kworker/u64:0-e 0.1
4514 speech-dispatch 0.1
2541 ibus-extension- 0.1
5312 kworker/u64:1-k 0.1
2047 ibus-daemon 0.1
1647 wpa_supplicant 0.1
1 systemd 0.1
2550 localsearch-3 0.0
1592 dbus-daemon 0.0
3916 Isolated Web Co 0.0
1763 wireplumber 0.0
1646 NetworkManager 0.0
18 rcu_preempt 0.0
2319 Xwayland 0.0
3019 WebExtensions 0.0
4355 Isolated Servic 0.0
4488 Web Content 0.0
4392 Web Content 0.0
2566 ibus-engine-moz 0.0
4528 sd_espeak-ng 0.0
2230 evolution-alarm 0.0
379 usb-storage 0.0
32 migration/2 0.0
2023 evolution-sourc 0.0
38 migration/3 0.0
2574 mozc_server 0.0
44 migration/4 0.0
1758 dbus-daemon 0.0
56 migration/6 0.0
68 migration/8 0.0
62 migration/7 0.0
1736 systemd 0.0
50 migration/5 0.0
620 systemd-journal 0.0
80 migration/10 0.0
26 migration/1 0.0
1597 polkitd 0.0
348 kworker/15:2-mm 0.0
92 migration/12 0.0
110 migration/15 0.0
4535 sd_espeak-ng 0.0
86 migration/11 0.0
```



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4535	sd_espeak-ng	0.0
86	migration/11	0.0
104	migration/14	0.0
2551	xdg-desktop-por	0.0
1978	colord	0.0
1603	udisksd	0.0
98	migration/13	0.0
138	khugepaged	0.0
74	migration/9	0.0
2410	evolution-addre	0.0
2055	gsd-media-keys	0.0
2046	upowerd	0.0
674	systemd-udev	0.0
2470	mutter-x11-fram	0.0
111	ksoftirqd/15	0.0
2056	gsd-power	0.0
21	migration/0	0.0
1681	ModemManager	0.0
2438	gsd-xsettings	0.0
903	kworker/u65:2-t	0.0
5304	kworker/1:0-eve	0.0
2664	kworker/u65:15-	0.0
2435	xdg-desktop-por	0.0
2392	evolution-calen	0.0
2073	gsd-wacom	0.0
1602	systemd-logind	0.0
52	kworker/5:0-eve	0.0
357	irq/62-PNP0C50:	0.0
2053	gsd-keyboard	0.0
4532	sd_dummy	0.0
2049	gsd-color	0.0
1861	gnome-session-b	0.0
1787	gnome-session-b	0.0
2943	Socket Process	0.0
4527	sd_espeak-ng-mb	0.0
2623	xdg-desktop-por	0.0
2065	gsd-sharing	0.0
2012	gnome-shell-cal	0.0
1722	kworker/9:2-eve	0.0
1589	bluetoothd	0.0
1766	gnome-keyring-d	0.0
1588	avahi-daemon	0.0
408	kworker/u65:1-t	0.0
2655	kworker/u65:6-t	0.0
2052	gsd-housekeepin	0.0
1598	smartd	0.0
1587	accounts-daemon	0.0
376	kworker/6:2-mm_	0.0
371	kworker/13:2-mm	0.0
51	ksoftirqd/5	0.0
2958	power-profiles-	0.0
2658	kworker/u65:9-t	0.0
2128	gjs	0.0
1707	cupsd	0.0
4638	bash	0.0
178	kworker/u65:0-r	0.0
27	ksoftirqd/1	0.0
10	kworker/0:1-mm_	0.0
2308	gvfs-afc-volume	0.0
2082	gvfs-udisks2-vo	0.0
2027	gjs	0.0
1945	cupd-browsed	0.0

1945	cupd-browsed	0.0
1776	rtkit-daemon	0.0
1762	pipewire	0.0
1599	switcheroo-cont	0.0
885	kworker/11:2-ev	0.0
365	kworker/1:2-eve	0.0
256	kworker/7:1-eve	0.0
4987	kworker/3:0-mm_	0.0
2657	kworker/u65:8-t	0.0
2660	kworker/u65:11-	0.0
2306	goa-identity-se	0.0
2296	goa-daemon	0.0
2068	gsd-sound	0.0
1851	gvfsd	0.0
1767	mpiris-proxy	0.0
1728	gdm-session-wor	0.0
1721	gdm3	0.0
1596	low-memory-moni	0.0
905	kworker/8:2-mm_	0.0
247	kworker/1:1H-kb	0.0
63	ksoftirqd/7	0.0
3298	ibus-engine-sim	0.0
2891	kworker/12:2-mm	0.0
2656	kworker/u65:7-r	0.0
2584	ibus-x11	0.0
2446	xdg-document-po	0.0
2204	gvfs-gphoto2-vo	0.0
2067	gsd-smartcard	0.0
2048	gsd-ally-settin	0.0
2060	gsd-rfkill	0.0
1840	ssh-agent	0.0
358	kworker/10:1-mm	0.0
286	kworker/5:1H-kb	0.0
136	kcompactd0	0.0
75	ksoftirqd/9	0.0
99	ksoftirqd/13	0.0
57	ksoftirqd/6	0.0
4853	kworker/2:0-eve	0.0
2652	kworker/u65:3-t	0.0
2659	kworker/u65:10-	0.0
2663	kworker/u65:14-	0.0
2403	gvfsd-metadata	0.0
2285	gvfs-goa-volume	0.0
2209	gsd-disk-utilit	0.0
2173	gvfs-mtp-volume	0.0
2070	gsd-usb-protect	0.0
2051	gsd-datetime	0.0
2058	gsd-print-notif	0.0
2064	gsd-screensaver	0.0
2022	dconf-service	0.0
1939	xdg-permission-	0.0
1908	dbus-daemon	0.0
1858	gvfsd-fuse	0.0
1782	gdm-wayland-ses	0.0
1591	cron	0.0
374	kworker/12:1-ev	0.0
361	kworker/2:4-eve	0.0
317	kworker/9:1H-kb	0.0
310	kworker/15:1H-k	0.0
166	kworker/4:1-eve	0.0
149	kworker/0:1H-kb	0.0
69	ksoftirqd/8	0.0





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80	ksoftirqd/0	0.0
81	ksoftirqd/10	0.0
93	ksoftirqd/12	0.0
33	ksoftirqd/2	0.0
39	ksoftirqd/3	0.0
45	ksoftirqd/4	0.0
17	ksoftirqd/0	0.0
2	kthread	0.0
3	pool_workqueue	0.0
4	kworker/R-kvfre	0.0
5	kworker/R-rcu_g	0.0
6	kworker/R-sync	0.0
7	kworker/R-slub	0.0
8	kworker/R-netns	0.0
11	kworker/0:0H-ev	0.0
13	kworker/R-mm_pe	0.0
14	rcu_tasks_kthre	0.0
15	rcu_tasks_rude	0.0
16	rcu_tasks_trace	0.0
19	rcu_exp_par_gp	0.0
20	rcu_exp_gp_kthr	0.0
22	idle_inject/0	0.0
23	cpuhp/0	0.0
24	cpuhp/1	0.0
25	idle_inject/1	0.0
29	kworker/1:0H-ev	0.0
30	cpuhp/2	0.0
31	idle_inject/2	0.0
35	kworker/2:0H-ev	0.0
36	cpuhp/3	0.0
37	idle_inject/3	0.0
41	kworker/3:0H-ev	0.0
42	cpuhp/4	0.0
43	idle_inject/4	0.0
47	kworker/4:0H-ev	0.0
48	cpuhp/5	0.0
49	idle_inject/5	0.0
53	kworker/5:0H-ev	0.0
54	cpuhp/6	0.0
55	idle_inject/6	0.0
59	kworker/6:0H-ev	0.0
60	cpuhp/7	0.0
61	idle_inject/7	0.0
65	kworker/7:0H-ev	0.0
66	cpuhp/8	0.0
67	idle_inject/8	0.0
70	kworker/8:0-slu	0.0
71	kworker/8:0H-ev	0.0
72	cpuhp/9	0.0
73	idle_inject/9	0.0
77	kworker/9:0H-ev	0.0
78	cpuhp/10	0.0
79	idle_inject/10	0.0
83	kworker/10:0H-e	0.0
84	cpuhp/11	0.0
85	idle_inject/11	0.0
87	ksoftirqd/11	0.0
89	kworker/11:0H-e	0.0
90	cpuhp/12	0.0
91	idle_inject/12	0.0
95	kworker/12:0H-e	0.0
96	cpuhp/13	0.0
97	idle_inject/13	0.0

101	kworker/13:0H-e	0.0
102	cpuhp/14	0.0
103	idle_inject/14	0.0
105	ksoftirqd/14	0.0
106	kworker/14:0-mm	0.0
107	kworker/14:0H-e	0.0
108	cpuhp/15	0.0
109	idle_inject/15	0.0
113	kworker/15:0H-e	0.0
130	kdevtmpfs	0.0
131	kworker/R-inet	0.0
132	kauditd	0.0
133	khungtaskd	0.0
134	oom_reaper	0.0
135	kworker/R-write	0.0
137	ksmd	0.0
139	kworker/R-kinte	0.0
140	kworker/R-kbloc	0.0
141	kworker/R-blkcg	0.0
142	irq/9-acpi	0.0
145	kworker/R-tpm_d	0.0
146	kworker/R-edac	0.0
147	kworker/R-devfr	0.0
148	kworker/R-quota	0.0
150	irq/29-AMD-Vi	0.0
151	kswapd0	0.0
153	kworker/15:1-ev	0.0
160	kworker/R-kthro	0.0
165	kworker/R-acpi	0.0
167	hwrng	0.0
169	kworker/R-mld	0.0
170	kworker/R-ipv6	0.0
173	kworker/4:1H-kb	0.0
176	kworker/R-kstrp	0.0
246	irq/30-ACPI:Eve	0.0
248	irq/31-ACPI:Eve	0.0
249	irq/32-ACPI:Eve	0.0
250	irq/33-ACPI:Eve	0.0
251	irq/34-ACPI:Eve	0.0
252	kworker/3:1-mm	0.0
253	irq/35-ACPI:Eve	0.0
254	kworker/9:1-eve	0.0
255	kworker/6:1-cgr	0.0
285	kworker/2:1H-kb	0.0
297	kworker/13:1H-k	0.0
309	kworker/7:1H-kb	0.0
311	kworker/14:1H-k	0.0
312	kworker/8:1H-kb	0.0
313	kworker/6:1H-kb	0.0
314	kworker/3:1H-kb	0.0
315	kworker/12:1H-k	0.0
316	kworker/10:1H-k	0.0
319	kworker/11:1H-k	0.0
351	kworker/R-nvme	0.0
352	kworker/R-nvme	0.0
353	kworker/R-nvme	0.0
354	kworker/R-nvme	0.0
355	watchdogd	0.0
372	kworker/R-amdgp	0.0
373	kworker/R-ttm	0.0
377	scsi_ah_0	0.0
378	kworker/R-scsi	0.0





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```
377 kworker/R-scsi_ 0.0
378 kworker/R-scsi_ 0.0
380 kworker/R-uas_ 0.0
381 kworker/R-amdgp 0.0
382 kworker/R-amdgp 0.0
383 kworker/R-amdgp 0.0
384 kworker/R-amdgp 0.0
385 kworker/R-amdgp 0.0
386 kworker/R-amdgp 0.0
387 kworker/R-amdgp 0.0
388 kworker/R-amdgp 0.0
389 kworker/R-amdgp 0.0
390 kworker/R-dm_vb 0.0
391 kworker/R-amdgp 0.0
392 card0-crtc0 0.0
393 card0-crtc1 0.0
394 card0-crtc2 0.0
395 card0-crtc3 0.0
396 kworker/R-gfx_0 0.0
397 kworker/R-comp_ 0.0
398 kworker/R-comp_ 0.0
399 kworker/R-comp_ 0.0
400 kworker/R-comp_ 0.0
401 kworker/R-comp_ 0.0
402 kworker/R-comp_ 0.0
403 kworker/R-comp_ 0.0
404 kworker/R-comp_ 0.0
405 kworker/R-sdma0 0.0
406 kworker/R-vcn_u 0.0
407 kworker/R-jpeg_ 0.0
417 kworker/R-md 0.0
418 kworker/R-md_bi 0.0
424 kworker/R-raid5 0.0
606 psimon 0.0
679 psimon 0.0
733 kworker/10:2-ev 0.0
760 kworker/R-crypt 0.0
878 kworker/R-cfg80 0.0
902 kworker/R-kmems 0.0
1601 avahi-daemon 0.0
1655 psimon 0.0
1690 krfcomm 0.0
1738 (sd-pam) 0.0
1838 gcr-ssh-agent 0.0
1839 gnome-session-c 0.0
1891 at-spi-bus-laun 0.0
1977 at-spi2-registr 0.0
2166 gsd-printer 0.0
2454 fusermount3 0.0
2540 ibus-dconf 0.0
2548 ibus-portal 0.0
2653 kworker/u65:4-t 0.0
2654 kworker/u65:5-t 0.0
2661 kworker/u65:12- 0.0
2662 kworker/u65:13- 0.0
2869 crashhelper 0.0
4751 kworker/1:1-mm_ 0.0
4834 kworker/7:0-eve 0.0
4866 kworker/13:0 0.0
5163 kworker/14:1-ev 0.0
5194 kworker/11:0 0.0
5264 kworker/5:2-eve 0.0
5292 kworker/0:0 0.0
5317 kworker/4:0-eve 0.0
5703 kworker/u65:16 0.0
5722 kworker/5:1-mm_ 0.0
5754 kworker/7:2-eve 0.0
5763 kworker/14:2-mm 0.0
5783 kworker/2:1-mm_ 0.0
5869 kworker/4:2-mm_ 0.0
5906 kworker/9:0 0.0
6002 ps 0.0
```

```
5163 kworker/14:1-ev 0.0
5194 kworker/11:0 0.0
5264 kworker/5:2-eve 0.0
5292 kworker/0:0 0.0
5317 kworker/4:0-eve 0.0
5703 kworker/u65:16 0.0
5722 kworker/5:1-mm_ 0.0
5754 kworker/7:2-eve 0.0
5763 kworker/14:2-mm 0.0
5783 kworker/2:1-mm_ 0.0
5869 kworker/4:2-mm_ 0.0
5906 kworker/9:0 0.0
6002 ps 0.0
```

user@debian:~\$





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CONCLUSION

A lo largo de la práctica, conocimos diferentes formas de utilizar el comando ps para consultar los procesos de Linux. Observamos que es posible obtener información general, consultar detalles específicos, filtrar resultados y ordenar los procesos según el uso de los recursos del sistema.

El uso de estas opciones nos permitió comprender mejor cómo se organiza la información de los procesos y cómo podemos utilizar la terminal para realizar consultas de manera más precisa. Cada comando tiene una función particular, por lo que elegir la opción adecuada depende de la información que necesitemos obtener.

